DSA CLASS TEST

Revised Set · 6 Story-Based Coding Questions | Mathematics, Arrays, Sorting, Strings, Hashing & Fundamentals

Suggested Duration	Total Marks	Language
90 minutes	60 (10 per question)	C++

Instructions: Write a complete solution for each question. Handle the given constraints and edge cases. Return the requested result without extra output.

Question 1: The Missing Seat Number

Topic: Mathematics | Difficulty: Easy

A college is organizing a programming competition. Each participant receives a unique seat number from `0` to n. However, one seat number is missing from the registration list.

You are given an array containing n distinct numbers from the range `0` to n. Find and return the missing number.

Example 1

```
Input: nums = [3, 0, 1]
Output: 2
```

Explanation: The numbers should range from 0 to 3. Since 2 is missing, return 2.

Example 2

```
Input: nums = [0, 1]
Output: 2
```

Constraints

1 <= nums.length <= 10^4 • All elements are distinct. • 0 <= nums[i] <= n

Question 2: Find the Best Pair of Students

Topic: Arrays | Difficulty: Easy

During a college event, students collect points by participating in activities. The organizer wants to find two students whose points add up to a given target.

You are given an integer array `scores` and an integer `target`. Return the indices of the two elements whose sum equals the target. Exactly one valid answer exists, and you cannot use the same element twice.

Example 1

```
Input: scores = [2, 7, 11, 15]; target = 9
Output: [0, 1]
```

Explanation: The scores at indices 0 and 1 are 2 and 7. Their sum is 9.

Example 2

```
Input: scores = [3, 2, 4]; target = 6
Output: [1, 2]
```

Constraints

2 <= scores.length <= 10^4 • -10^9 <= scores[i] <= 10^9 • Exactly one valid answer exists.

Question 3: Find the Second-Highest Score

Topic: Sorting / Arrays | Difficulty: Easy–Medium

After a coding contest, a teacher sorts the students' scores to announce the runner-up. Some students may have the same score, so the runner-up must have a score strictly lower than the highest score.

Given an integer array `scores`, return the second-largest **distinct** value. If a second distinct value does not exist, return `-1`.

Example 1

```
Input: scores = [8, 3, 5, 8, 6]
Output: 6
```

Explanation: The highest distinct score is 8, and the second-highest distinct score is 6.

Example 2

```
Input: scores = [4, 4, 4]
Output: -1
```

Explanation: There is no second distinct score.

Constraints

1 <= scores.length <= 10^4 • -10^5 <= scores[i] <= 10^5

Question 4: Find the First Unique Letter

Topic: Strings | Difficulty: Easy–Medium

A teacher creates a short access code from letters. To make it easier to identify, the teacher wants to find the first letter that appears exactly once in the code.

Given a string `code` containing lowercase English letters, return the index of the first character that appears only once. If every character repeats, return `-1`.

Example 1

```
Input: code = "leetcode"
Output: 0
```

Explanation: The first character, "l", appears only once.

Example 2

```
Input: code = "loveleetcode"
Output: 2
```

Explanation: The character "v" is the first character that appears only once.

Example 3

```
Input: code = "aabb"
Output: -1
```

Explanation: Every character appears more than once.

Constraints

1 <= code.length <= 10^5 • code contains only lowercase English letters.

Question 5: Find the First Repeated Roll Number

Topic: Hashing | Difficulty: Easy

A teacher records the roll numbers of students who submitted their assignments. Due to a mistake, some roll numbers may have been entered more than once.

You are given an integer array `rollNumbers`. Return `true` if any roll number appears at least twice; otherwise, return `false`.

Example 1

```
Input: rollNumbers = [1, 2, 3, 1]
Output: true
```

Explanation: Roll number 1 appears twice.

Example 2

```
Input: rollNumbers = [1, 2, 3, 4]
Output: false
```

Explanation: Every roll number is unique.

Example 3

```
Input: rollNumbers = [1, 1, 1, 3, 3, 4, 3, 2, 4, 2]
Output: true
```

Constraints

1 <= rollNumbers.length <= 10^5 • -10^9 <= rollNumbers[i] <= 10^9

Question 6: The Best Buy-and-Sell Day

Topic: Array Traversal / Fundamentals | Difficulty: Easy–Medium

A student tracks the price of a study-planner app over several days. The student wants to buy the app at its lowest price on one day and sell it at a higher price on a later day.

You are given an array `prices`, where `prices[i]` is the price on day `i`. Find the maximum profit possible by buying on one day and selling on a later day. If no profit is possible, return `0`.

Example 1

```
Input: prices = [7, 1, 5, 3, 6, 4]
Output: 5
```

Explanation: Buy at price 1 and sell at price 6. The maximum profit is 6 - 1 = 5.

Example 2

```
Input: prices = [7, 6, 4, 3, 1]
Output: 0
```

Explanation: The price keeps decreasing, so no profitable transaction is possible.

Constraints

1 <= prices.length <= 10^5 • 0 <= prices[i] <= 10^4 • You must buy before selling.

Teacher's Answer Key

Keep this section separate from the student question paper.

Q.	Concept	Expected approach	Target complexity
1	Mathematics	Sum formula or XOR	O(n)
2	Arrays	Hash map or two loops	O(n) expected with hashing
3	Sorting / Arrays	Track two distinct maxima or sort and skip duplicates	O(n) or O(n log n)
4	Strings	Count letters, then scan from left to right	O(n)
5	Hashing	unordered_set	O(n) expected
6	Array traversal	Track minimum price and maximum profit	O(n)

Suggested evaluation: 10 marks per question — 6 for correct logic, 2 for edge cases, and 2 for complexity and code quality.

Difficulty note: Questions 3 and 4 have been replaced with slightly more challenging problems while remaining suitable for a basic-to-low-medium class test.